

Lian Cheng Guo

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EDUCATION AND CERTIFICATIONS

University of Waterloo

Waterloo, ON

Candidate for Bachelor of Applied Science in Computer Engineering, Honours, Co-op

Sept. 2022 – Present

Relevant Courses: Computational Intelligence, Algorithms & Data Structures, Database Systems, Systems

Programming & Concurrency, Real-Time Operating Systems

Deep Learning Specialization

DeepLearning.AI on Coursera

Mar 2025

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, HTML, CSS, JavaScript, TypeScript

Frameworks & Libraries: TensorFlow, Keras, LangChain, React, Next.js, Three.js, Tailwind CSS, pytest, Selenium

Tools & Technologies: Azure AI Services, PostgreSQL, MySQL, Supabase, Chroma, Ollama, Git, GitHub, VS Code

EXPERIENCE

AI Software Engineer Intern

Jan 2026 – May 2026

Correctional Service Canada

Ottawa, ON

- Contributed to a **RAG-based AI chatbot** web application using Azure services to deliver document-grounded responses, reducing frontline document lookup from **minutes to seconds**, accelerating workflows by up to **100x**.
- Developed backend automation for document metadata tagging, storage updates, and indexing workflows, reducing manual update time from **over 2 hours to under 5 minutes** while enabling metadata-based retrieval filtering.
- Optimized RAG prompt and retrieval behavior by analyzing token usage, evaluating chunking strategies, and validating response quality through targeted test scenarios, achieving a **95%+ expected-behavior pass rate**.
- Improved chatbot front-end functionality by updating HTML, CSS, and JavaScript components, integrating API calls, and refining the user interface for better usability.

AI Software Engineer Intern

Jan 2025 – Apr 2025

Sofp Inc.

Richmond Hill, ON

- Implemented **Transformer model components** in Python using TensorFlow and Keras, including positional encodings, encoder-decoder blocks, multi-head attention, and softmax-based sequence prediction.
- Developed a **YOLOv2-based vehicle detection pipeline** with filtering, non-max suppression, anchor boxes, and IoU evaluation to validate object detection performance on test images.
- Prepared large-scale datasets for training, fine-tuning, and evaluation workflows, improving data consistency for language model and computer vision experiments.
- Built an **LLM knowledge base** using Ollama, LangChain, and Chroma, enabling retrieval-based Q&A experimentation and improving training efficiency by **30%**.

Software QA Intern

Sept 2023 – Dec 2023

Interaptix Augmented Reality

Toronto, ON

- Developed and maintained automated UI and regression tests using pytest and Selenium, validating **20+ test scenarios** and reducing manual QA effort by over **50%**.
- Validated full-stack functionality across web, iOS, and AR workflows, covering **100+ regression test cases** and reducing release-blocking compatibility issues.
- Collaborated with developers and product designers to validate **3 major releases** and **4 hotfixes**, reducing post-release defects by **25%** through structured test plans and issue tracking.

PROJECTS

Interactive Rubik's Cube Solver 🐼 | Next.js, TypeScript, Tailwind CSS, Supabase, OpenAI API

Apr 2026

- Built an interactive Rubik's Cube solver web app with manual face-turn controls, scramble/reset functionality, solved-state detection, and step-by-step solution playback.
- Implemented cube state management and move tracking to maintain valid state transitions, record solve history, and compute structured metrics from user move sequences.
- Developed backend route handlers and Supabase integration to persist solve attempts, including scramble data, move counts, timestamps, and analysis results.
- Added AI-generated coaching feedback grounded in deterministic solve analysis, highlighting inefficient patterns such as redundant turns and inverse move pairs to help users improve solve efficiency.